

REC Guardian: Pre-Configured Fraud Scenarios & Testing Guide

Purpose: This guide details the 5 realistic pre-configured forensic scenarios seeded into the platform database, explaining the fraud mechanics, which detection engines trigger, and how regulators investigate and adjudicate them.



Scenario Matrix Overview

Scenario #	Scenario Description	Target Identifier	Primary Engine Flags	Risk Score	Expected Outcome
1	Clean Legitimate Claim	CLM-2026-LEGIT-01	None (Within 2% meter tolerance, normal CF)	14.2 / 100	Auto-Approved, REC Minted (REC-2026-SOL-09921)
2	Meter Mismatch Over-Claim	CLM-2026-FRAUD-MTR	RULE-005 (Meter Mismatch +216.7%), ML Outlier	88.0 / 100	Status HELD , Case Opened (CASE-2026-MTR-001)
3	Physical Capacity Impossibility	CLM-2026-FRAUD-CAP	RULE-003 & RULE-004 (Exceeds Physical Max by >250%)	95.0 / 100	Status HELD , Case Opened (CASE-2026-CAP-001)
4	Document Evidence Reuse	CLM-2026-FRAUD-DOC	RULE-009 (Exact SHA-256 PDF report collision)	89.5 / 100	Status HELD , Cross-Facility Alert
5	Circular Wash Trading Ring	REC-2026-WND-88319	NetworkX Directed Graph Cycle Detection ($A \rightarrow B \rightarrow C \rightarrow A$)	Graph Alert	Flagged in Graph Surveillance & Lineage

1. Scenario 1: Clean Legitimate Generation

Fraud Anatomy

- Facility:** Mojave Desert Solar One (50 MW Solar PV)
- Claimed Volume:** 1,200.0 MWh over 31-day period (March 2026)
- Substation Meter Reading:** 1,195.0 MWh Net Generation
- Capacity Factor:** $\frac{1200}{50 \times 744} \approx 3.2\%$ (Early commissioning stage, within valid baseline)

Detection Pipeline Evaluation

- Deterministic Rules:**
- RULE-005** (Meter Mismatch): Difference is $\frac{1200-1195}{1195} \approx 0.41\% \leq 2.0\%$ tolerance $\implies$ **PASSED**.

- **RULE-003** (Capacity Bounds): Well within 50 MW physical envelope $\implies$ **PASSED**.
- **RULE-008** (Duplicate Check): Unique fingerprint $\implies$ **PASSED**.
- **ML Anomaly Model**: Isolation Forest classifies vector as inlier $\implies$ Anomaly Score: **12.0 / 100**.
- **Risk Fusion**: Composite score **14.2 / 100** (Recommendation: **APPROVE**).

System Action

1. Claim automatically transitions to **APPROVED** .
2. Digital REC certificate minted onto ledger: **REC-2026-SOL-09921** (Volume: 1,200 MWh, Vintage: 2026-03).
3. Cryptographic block committed to SHA-256 ledger.

2. Scenario 2: Smart Meter vs. Claim Mismatch (+216% Over-Claim)

Fraud Anatomy

- **Facility**: Mojave Desert Solar One (50 MW Solar PV)
- **Substation Calibrated Meter**: **1,200.0 MWh**
- **Producer Claimed Volume**: **3,800.0 MWh**
- **Over-Claim Delta**: +2,600.0 MWh (+216.7% inflation above actual generation)

Detection Pipeline Evaluation

- **Deterministic Rule Engine**:
 - **RULE-005** (**METER_CLAIM_MISMATCH**) triggers with **CRITICAL** severity.
 - Formula: $\Delta = \frac{3800-1200}{1200} = +216.7\% \gg 2.0\%$.
 - Rule engine score contribution: **+80.0 points**.
- **ML Anomaly Model**:
 - Input vector $\vec{X} = [0.102, 3.16, 76.0]$ severely deviates from calibrated solar baselines.
 - Isolation Forest flags extreme statistical outlier: Anomaly score **92.4 / 100**.
- **Risk Fusion**:
 - Score reaches **88.0 / 100** (Risk Level: **CRITICAL** , Recommendation: **HOLD**).

System Action & Regulatory Investigation

1. Claim is automatically locked into status **HELD** . Certificate issuance is blocked.
2. An investigation docket is auto-opened: **CASE-2026-MTR-001** (Priority: **CRITICAL**).
3. Case appears in the Regulatory Authority console for human adjudication (**CONFIRM_FRAUD_HOLD** vs **CLEAR_AND_ISSUE**).

3. Scenario 3: Physical Capacity Impossibility (>250% Above Physics)

Fraud Anatomy

- **Facility**: Mojave Desert Solar One (50 MW Nameplate Capacity)
- **Interval**: 1 Month (720 Hours)

- **Theoretical Absolute Physical Limit:**

$$\text{Max Possible Energy} = 50 \text{ MW} \times 720 \text{ Hours} = 36,000 \text{ MWh}$$

(Even if running at 100% capacity factor 24 hours a day with zero night darkness)

- **Claimed Volume: 90,000.0 MWh** ($> 250\%$ above theoretical physical limits!)

Detection Pipeline Evaluation

- **Deterministic Rule Engine:**
- **RULE-004** (**THEORETICAL_CAPACITY_LIMIT**) triggers: Claimed 90,000 MWh exceeds the absolute mathematical ceiling of 36,000 MWh.
- **RULE-003** (**NAMEPLATE_CAPACITY_VIOLATION**) triggers: Capacity factor is **250.0%** (Solar maximum realistic CF is $\leq 32\%$).
- **Risk Fusion Hard Override:**
- Because a physical impossibility was detected, the hard override activates:

$$\text{FinalScore} = \max(\text{FinalScore}, 95.0) = 95.0/100$$

System Action

1. Claim immediately flagged **CRITICAL** and held.
2. Investigation Case **CASE-2026-CAP-001** opened with findings: *"Physical impossibility: Generation exceeds 100% continuous nameplate capacity by 2.5x."*

4. Scenario 4: Cross-Facility Document Evidence Hash Reuse

Fraud Anatomy

- **Legitimate Origin:** Mojave Desert Solar One uploaded third-party engineering audit report **mojave_solar_audit_q1_2026.pdf** with SHA-256 hash: **e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855**
- **Fraudulent Submitter:** Columbia Gorge Wind Facility submits an unrelated wind generation claim, attaching the exact same PDF audit report.

Detection Pipeline Evaluation

- **Deterministic Rule Engine:**
- **RULE-009** (**DOCUMENT_EVIDENCE_REUSE**) computes the SHA-256 checksum of the evidence file.
- Cross-references historical registry attachments: Detects that hash **e3b0c442 ...** was previously registered to a completely different facility (**Plant #1** vs **Plant #3**).
- Score contribution: **+85.0 points**.

System Action

1. Claim blocked from minting.
 2. Forensic audit logs document recycling alert across facility boundaries.
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5. Scenario 5: Circular Wash Trading Ring ($A \rightarrow B \rightarrow C \rightarrow A$)

Fraud Anatomy

- **Certificate:** REC-2026-WND-88319 (5,000 MWh Wind REC)
- **Trade Hop 1:** Boreas Wind Energy (user-5) transfers to Global Carbon Exchange (user-6).
- **Trade Hop 2:** Global Carbon Exchange (user-6) transfers to Helios Solar (user-4).
- **Trade Hop 3:** Helios Solar (user-4) transfers *back* to Boreas Wind Energy (user-5).
- **Objective:** Artificial volume pump / phantom liquidity laundering without real carbon offset transfer.

Detection Pipeline Evaluation

- **Graph Relationship Engine:**
- Builds directed trading graph from CertificateTransfer records.
- Runs Johnson's elementary cycle algorithm:

Cycle Detected : user-5 $\longrightarrow$ user-6 $\longrightarrow$ user-4 $\longrightarrow$ user-5

- Computes clustering coefficient and marks counterparties as wash-trading ring participants.

System Action

1. Visual Network Graph in the UI highlights the circular cycle edges in bright crimson.
2. Certificate Lineage Explorer displays all 3 secondary trading hops with transaction hashes.

6. How to Test & Demo These Scenarios

Step 1: Trace Any Scenario in the Lineage Explorer

1. Navigate to the **Certificate Explorer & Timeline** tab in the UI.
2. Click any of the pre-configured preset buttons:
3. Meter Overclaim (+216%) $\implies$ loads CLM-2026-FRAUD-MTR
4. Capacity Impossible (>240%) $\implies$ loads CLM-2026-FRAUD-CAP
5. Circular Wash Ring (3 Hops) $\implies$ loads REC-2026-WND-88319
6. Observe the full 6-stage lineage card and the step-by-step chronological **Evidence Timeline**.

Step 2: Execute Regulatory Adjudication

1. Log in as **Regulatory Officer** (regulator@recguardian.org).
2. Open the **Regulatory Cases** tab.
3. Click "**Adjudicate Case**" on CASE-2026-MTR-001 .
4. Choose "**Confirm Fraud Hold**", enter regulatory justification notes, and click "**Sign & Seal Decision**".
5. Observe the claim status permanently sealed as rejected and the resolution block committed to the ledger.